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Antibody Responses to SARS-CoV-2 in Patients With Novel Coronavirus Disease 2019

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Background. The novel coronavirus SARS-CoV-2 is a newly emerging virus. The antibody response in infected patients remains largely unknown, and the clinical value of antibody testing has not been fully demonstrated.

Methods. 173 patients with SARS-CoV-2 infection were enrolled. Their serial plasma samples (n = 535) collected during hospitalization were tested for total antibodies (Ab), IgM, and IgG against SARS-CoV-2. The dynamics of antibodies with disease progress were analyzed.

Results. Among 173 patients, the seroconversion rates for Ab, IgM, and IgG were 93.1%, 82.7%, and 64.7%, respectively. The reason for the negative antibody findings in 12 patients might be due to the lack of blood samples at the later stage of illness. The median seroconversion times for Ab, IgM, and then IgG were days 11, 12, and 4, respectively. The presence of antibodies was <40% among patients within 1 week of onset, and rapidly increased to 100.0% (Ab), 94.3% (IgM), and 79.8% (IgG) by day 15 after onset. In contrast, RNA detectability decreased from 66.7% (58/87) in samples collected before day 7 to 45.5% (25/55) during days 15–39. Combining RNA and antibody detection significantly improved the sensitivity of pathogenic diagnosis for COVID-19 ($P < .001$), even in the early phase of 1 week from onset ($P = .007$). Moreover, a higher titer of Ab was independently associated with a worse clinical classification ($P = .006$).

Conclusions. Antibody detection offers vital clinical information during the course of SARS-CoV-2 infection. The findings provide strong empirical support for the routine application of serological testing in the diagnosis and management of COVID-19 patients.

Keywords. antibody; SARS-CoV-2; COVID-19.

Since early December of 2019 and up to 22 March 2020, over 260 000 cases of coronavirus disease 2019 (COVID-19) caused by the novel severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection, with over 11 000 deaths reported in 184 countries [1]. The World Health Organization declared COVID-19 as a pandemic on 11 March 2020. According to recent reports, most patients with COVID-19 have an incubation period of 3 to 7 days [2]. Fever, cough, and fatigue are the most common symptoms, whereas nasal congestion, runny nose, and diarrhea are only noted in a small proportion of patients [3]. Severe cases might rapidly progress to acute respiratory distress syndrome (ARDS), septic shock, and difficult-to-treat metabolic acidosis and bleeding and coagulation dysfunction [4]. It should be noted that some patients with COVID-19 only had mild atypical symptoms initially, even in severe and critical cases [5]. Chest computed tomography (CT) scans of patients with COVID-19 are characterized by ground-glass opacity and bilateral patchy shadowing [6]. For laboratory tests, it was reported that most patients had lymphopenia and elevated C-reactive protein [7]. However, the above-mentioned clinical and laboratory characteristics are not easily distinguishable from pneumonia induced by infection with other common respiratory tract pathogens.

The timely and accurate diagnosis of the SARS-CoV-2 infection is the cornerstone of efforts to provide appropriate treatment for patients, to limit further spread of the virus and ultimately to eliminate the virus from human society. Currently, polymerase chain reaction (PCR)–based viral RNA detection is almost the only way to confirm the diagnosis of SARS-CoV-2 infection in practice. On

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the other hand, RNA testing based on throat or nasopharyngeal swabs has shown negligible false-negative risk [8]. The reported rate of positive tests has varied for different swab specimens in patients with COVID-19 [3, 9]. Many cases that were strongly epidemiologically linked to SARS-CoV-2 exposure and with typical lung radiological findings were RNA negative in their upper respiratory tract samples. The performance of reverse transcriptase (RT)-PCR depends on many factors, such as the sample types [9], different stage of infection in patients [10], the skill of sample collection, and the quality and consistency of the PCR assays being used. These problems lead to a critical delay in early diagnosis and management and present a serious challenge to providing timely life-support treatment and preventive quarantine.

Compared with PCR, serological testing is advantageous with a faster turnaround time, high-throughput, and less workload. However, the clinical value of antibodies largely depends on the understanding of host antibody responses during infection. Given that SARS-CoV-2 is a newly emerging virus, the antibody response in patients with COVID-19 remains largely unknown. This study investigates the dynamics of total antibodies (Ab), and immunoglobulin (Ig) M and IgG antibodies against SARS-CoV-2 in serial blood samples collected from 173 patients with confirmed COVID-19 and provides a discussion on the clinical value of antibody testing.

METHODS

Patients

A confirmed COVID-19 case and the clinical classification were defined based on the *New Coronavirus Pneumonia Prevention and Control Program* (4th edition) published by the National Health Commission of China. Briefly, patients with acute respiratory infection syndromes and/or abnormalities on chest CT images accompanied by detectable SARS-CoV-2 RNA in respiratory samples since illness onset at least once were confirmed to be a COVID-19 case. Patients were classified as being in a critical illness condition with the presence of ARDS or oxygen saturation less than 93% and who required mechanical ventilation either invasively or noninvasively. This study enrolled a total of 173 patients with COVID-19 who were admitted to the Shenzhen Third People's Hospital between 11 Janurary and 9 February 2020, and who were willing to donate their blood samples. All enrolled cases were confirmed to be infected with SARS-CoV-2 by use of real-time RT-PCR on samples from the respiratory tract. For all enrolled patients, the date of illness onset, clinical classification, RNA testing results during the hospitalization period, and personal demographic information were obtained from the clinical records. This study was reviewed and approved by the Medical Ethical Committee of Shenzhen Third People's Hospital (2020-0018). Written informed consent was obtained from each enrolled patient.

Antibody Measurement

The Ab, IgM antibody, and IgG antibody against SARS-CoV-2 in plasma samples were tested using enzyme-linked immunosorbent assay (ELISA) kits supplied by Beijing Wantai Biological Pharmacy Enterprise Co, Ltd, according to the manufacturer's instructions. Briefly, the ELISA for Ab detection was developed based on a double-antigen sandwich immunoassay (Ab-ELISA), using mammalian cell–expressed recombinant antigens containing the receptor binding domain (RBD) of the spike protein of SARS-CoV-2 as the immobilized and horseradish peroxidase–conjugated antigen. The IgM μ-chain capture method (IgM-ELISA) was used to detect the IgM antibodies, using the same HRP-conjugate RBD antigen as the Ab-ELISA. The IgG antibodies were tested using an indirect ELISA kit (IgG-ELISA) based on a recombinant nucleoprotein. The specificity of the assays for Ab, IgM, and IgG was determined to be 99.1% (211/213), 98.6% (210/213), and 99.0% (195/197) by testing of samples collected from healthy individuals before the outbreak of SARS-CoV-2.

Statistical Analysis

For continuous variables, means with SDs were used for normally distributed data and medians with interquartile ranges (IQRs) were used for non–normally distributed data. Cumulative seroconversion rates were calculated by using the Kaplan-Meier method. The association between antibody level and severity of illness was estimated by a generalized estimating equations (GEE) model with logit link function. All statistical analyses were conducted using SAS version 9.4 (SAS Institute, Cary, NC).

RESULTS

Characterization of Patients

Of a total of 368 patients with COVID-19 admitted to the hospital (before 9 February 2020), 173 (47%) were enrolled in the study (Table 1). The median age of the study patients was 48 years (IQR, 35–61 years) and 51.4% were female. There were 116 (67%) patients who had a clear epidemiological travel/residence history in Wuhan. Among them, 32 (18.5%) were in a critical illness condition. By 19 February, a total of 62 patients (35.8%) had recovered and were discharged from the hospital and 2 (1.1%) patients had died with underlying chronic disease.

Seroconversion of Antibodies Against SARS-CoV-2 in Patients With COVID-19

A total of 535 plasma samples collected during the hospitalization period of the 173 patients were tested for antibodies against SARS-CoV-2. The seroconversion rates for Ab, IgM, and IgG were 93.1% (161/173), 82.7% (143/173), and 64.7% (112/173), respectively (Table 1). Twelve patients who remained seronegative for Ab testing, possibly because their samples were all collected at the early stage of illness (10 samples earlier than day 10, the other 2 samples on days 11 and 13 after onset). The cumulative seroconversion curve showed that the rate for both Ab and IgM reached 100% at approximately 1 month after

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Table 1. Demographic and Clinical Characteristics of Patients and Sample Cohort With COVID-19 in This Study

	Total	Noncritical	Critical
Number	173	141	32
Gender, n (%)			
Male	84 (49)	63 (45)	21 (66)
Female	89 (51)	78 (55)	11 (34)
Age, median (IQR), years	48 (35–61)	41 (33–56)	64 (58–66)
Epidemiological exposure (1 month), n (%)			
Been to Wuhan	116 (67)	92 (65)	24 (75)
Unclear or others	57 (33)	49 (35)	8 (25)
Comorbidities,[a]n (%)	41 (24)	26 (18)	15 (47)
Clinical outcome, n (%)			
Recovery	62 (36)	54 (38)	8 (25)
Still in hospital	109 (63)	89 (62)	22 (69)
Death	2 (1.2)	0	2 (6.3)
RNA-confirmed time since onset, median (IQR),[b]days	4 (3–6)	4 (3–6)	6 (4–10)
Days since onset of first sample for antibody test, median (IQR)[c]	7 (5–10)	7 (5–9)	10 (6–16)
RNA (TS/NS) at the involved first sample, n (%)			
Positive	89 (51)	73 (52)	16 (50)
Negative	65 (38)	55 (39)	10 (31)
No data	19 (11)	13 (9.2)	6 (19)
rRT-PCR CT, median (IQR)	29 (25–31)	29 (24–32)	29 (28–31)
No. of antibody tested samples			
Of each case, median (IQR)	3 (2–4)	3 (2–4)	4 (3–5)
Total	535	404	131

Data are presented as medians (IQRs) and n (%).

Abbreviations: COVID-19, coronavirus disease 2019; CT, computed tomography; IQR, interquartile range; NS, nasal swabs; rRT-PCR, real-time reverse transcriptase–polymerase chain reaction; TS, throat swabs.

[a]Comorbidities included hypertension (n = 20), diabetes (n = 11), coronary heart disease (n = 3), chronic hepatitis B infection (n = 5), tumors (n = 2), obstructive sleep apnea syndrome (n = 1), chronic bronchitis (n = 1), hyperlipidemia (n = 1), renal insufficiency (n = 1), tuberculosis (cured, n = 1), and fatty liver disease (n = 1).

[b]The data indicated the time of confirmation for positive SARS-CoV-2 infection by using rRT-PCR on the respiratory sample since illness onset.

[c]The data indicated the time since illness onset of the first plasma sample of patients involved for serological tests in this study.

onset. Seroconversion sequentially occurred for Ab, IgM, and then IgG (Figure 1A). The median time to Ab, IgM, and IgG seroconversion was 11, 12, and 14 days, respectively. One of 2 patients tested on the onset day was seropositive. Overall, the seroconversion of Ab was significantly quicker than that of IgM ($P = .012$) and IgG ($P < .001$), possibly attributed to the double-antigen sandwich form of the assay used, which usually shows much higher sensitivity than the capture assay (IgM) and indirect assay (IgG). Moreover, all isotypes of virus-specific antibodies, including IgM, IgA, and IgG, can be detected by a double-sandwich assay, which may also contribute to the superior performance of the Ab test. In comparison to seroconversion rates of antibodies between critical and noncritical patients, none of the 3 markers showed a significant difference (data not shown).

Diagnostic Value of Antibody Assays for Patients at Different Times After Onset

In the early phase of illness within 7 day from onset, the RNA test had the highest sensitivity of 66.7%, whereas the antibody assays only had a positive rate of 38.3% (Table 2). However, the sensitivity of Ab overtook that of the RNA test from day 8 after onset and reached over 90% by day 12 after onset (Figure 1B). In samples from patients during days 8–14 after onset, the sensitivities of Ab (89.6%), IgM (73.3%), and IgG (54.1%) were all higher than that of the RNA test (54.0%) (Table 2). Among samples from patients in later phases (day 15–39 from onset), the sensitivities of Ab, IgM, and IgG were 100.0%, 94.3%, and 79.8%, respectively. In contrast, RNA was only detectable in 45.5% of samples on days 15–39. Further analyses demonstrated that, in patients with undetectable RNA in their respiratory tract samples collected during day 1–3, day 4–7, day 8–14 and day 15–39 from onset, 28.6% (2/7), 53.6% (15/28), 98.2% (56/57), and 100% (30/30) had detectable antibody in their Ab assays, respectively (Supplementary Table 1). Furthermore, combined use of the RNA and Ab test markedly improved the sensitivities of pathogenic diagnosis of COVID-19 in patients at different phases (Table 2).

Dynamics of Antibody Levels With the Progress and Severity of Disease

To investigate the dynamics of antibody levels according to disease course, the antibody levels were expressed using the relative binding signals compared with the cutoff value of each assay (S/CO). The longitudinal changes in antibody and RNA in 9 representative patients, including 6 in the noncritical group (Figure 2A) and 3 in the critical group (Figure 2B), are

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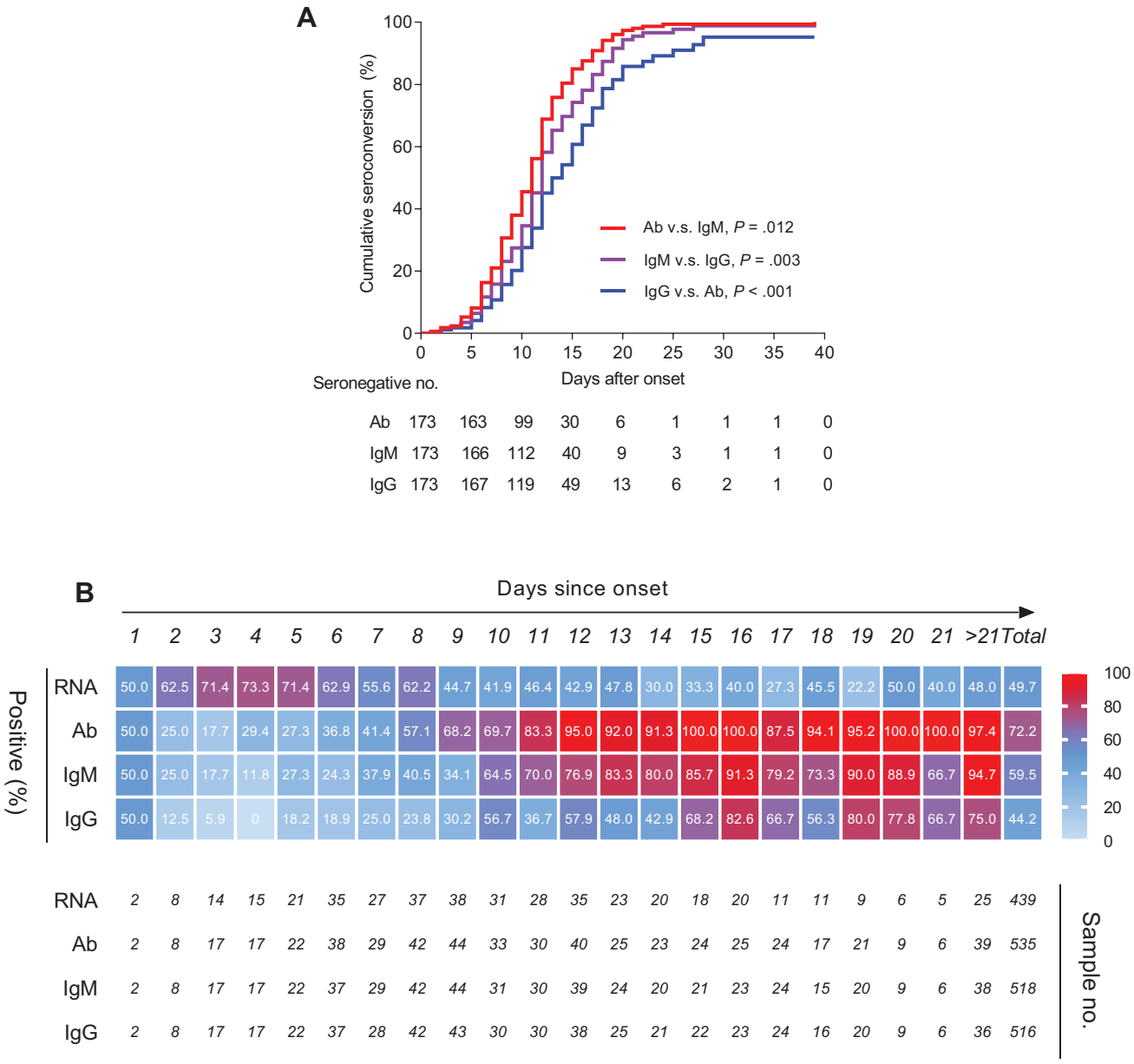


Figure 1. Cumulative incidence of seroconversion of antibodies against SARS-CoV-2 among patients with COVID-19 during the acute phase since illness onset. *A*, Cumulative incidence of seroconversion of Ab, IgM, and IgG among 173 patients of this study. *P* values were determined by log-rank test to compare different markers. *B*, Profiling of sensitivity performance of RNA, Ab, IgM, and IgG in time series since illness onset. A heat-map of detection of SARS-CoV-2 infection according to the time (days) since onset by a single RNA or antibody test is shown. Abbreviations: Ab, total antibodies; COVID-19, coronavirus disease 2019; Ig, immunoglobulin; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

presented in Figure 3. The first positive time point of RNA tests appeared earlier than that of the Ab test in 7 of 9 patients, except for case 185 (Ab was detectable 2 days earlier than RNA) and case 111 (on the same day). It should be noted that the increases in antibodies were not always accompanied by RNA clearance, particularly in the 3 critical patients. This finding suggested that antibodies may not be sufficient to clear the virus. In the pooled analyses on all involved patients, the average antibody levels showed a marked increase at approximately 1 week after onset and were continuously elevated during the next 2 weeks (Figure 3A). Further analyses suggested that there was no significant difference in the average S/CO value of Ab tests between critical and noncritical patients before day 12 after onset (Figure 3B). However, critical patients showed significantly higher Ab S/CO values than noncritical cases at approximately 2 weeks after onset ($P = .02$), and this association was not significant for either the IgM or IgG tests (data not shown). For further exploration, we determined the relative Ab titer of these samples (expressed as relative optical density) by serial dilution measurements of each sample. The quantitative data of Ab

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Table 2. Performance of Different Tests in Samples at Different Times Since Onset in Patients

Days After Onset	n	RNA		Ab		IgM		IgG		RNA + Ab	
		n(+)	Sensitivity, % (95% CI)	n(+)	Sensitivity, % (95% CI)	n(+)	Sensitivity, % (95% CI)	n(+)	Sensitivity, % (95% CI)	n(+)	Sensitivity, % (95% CI)
Total	173	112	67.1 (59.4, 74.1)	161	93.1 (88.2, 96.4)	143	82.7 (76.2, 88)	112	64.7 (57.1, 71.8)	172	99.4 (96.8, 100.0)
1–7	94	58[a]	66.7 (55.7, 76.4)	36	38.3 (28.5, 48.9)	27	28.7 (19.9, 39.0)	18	19.1 (11.8, 28.6)	74	78.7 (69.1, 86.5)
8–14	135	67[a]	54.0 (44.8, 63.0)	121	89.6 (83.2, 94.2)	99	73.3 (65.0, 80.6)	73	54.1 (45.3, 62.7)	131	97.0 (92.6, 99.2)
15–39	90	25[a]	45.5 (32.0, 59.5)	90	100.0 (96.0, 100.0)	83[b]	94.3 (87.2, 98.1)	71[c]	79.8 (69.9, 87.6)	90	100.0 (96.0, 100.0)

Abbreviations: Ab, total antibodies; CI, confidence interval; Ig, immunoglobulin; +, positive.

[a]There were 7, 11, and 35 patients who had not had RNA testing during days 1–7, 8–14, and 15–39 from onset, respectively.

[b]Two patients missed IgM tests due to inadequate plasma samples.

[c]One patient missed IgG tests due to inadequate plasma samples.

titers also revealed a significant difference ($P = .004$) between patients in the critical and noncritical groups (Figure 3C). Multivariate longitudinal GEE analyses suggested that age ($\beta = 0.139$, $P < .001$), gender ($\beta = 1.415$, $P = .006$), and Ab titer ($\beta = 0.336$, $P = .006$) were the independent factors strongly associated with the clinical classification based on the severity of illness (Supplementary Table 2).

DISCUSSION

The present data show that typical antibody responses to acute viral infection are wildly induced in patients with COVID-19. As expected, the Ab was first detected, followed by IgM and IgG. The seroconversion rate and the antibody levels increased rapidly during the first 2 weeks, the cumulative seropositive rate reached 50% on the 11th day and 100% on the 39th day. The seroconversion time of Ab and IgM and IgG antibodies appeared consecutively ($P < .05$), with a median seroconversion day of 11, 12, and 14, respectively. Due to the lack of blood samples collected from patients in the later stage of illness, how long the antibodies could last remains unknown. Our results demonstrate an excellent sensitivity of the Ab test in detection of patients' samples from 1 week after onset. Notably, even in the early stages of the illness within 1 week, some patients with undetectable RNA could be screened through Ab testing. Combining RNA and antibody tests significantly raised the sensitivity for detecting patients with SARS-CoV-2 ($P < .001$). These findings indicate that serological testing be an important supplement to RNA detection during the illness course.

To date, the confirmation of diagnosis of SARS-CoV-2 infection entirely depends on viral RNA testing. Even though it has high analytical sensitivity, the real-world performance of RNA test is not conclusive. Many suspected patients had to be tested for several days with multiple samples before a confirmed diagnosis was made, and during this waiting time they might not have been prioritized to receive relevant treatment and quarantine management [3]. These problems impede the timely diagnosis of SARS-CoV-2 infection, which is important for taking relevant actions to limit the damage of the current outbreak. Our study provided robust evidence that (1) the acute antibody response in patients with SARS-CoV-2 is very similar to many other acute viral infections, (2) serological testing can be a powerful approach in achieving timely diagnosis, and (3) the total antibody (Ab) test is more sensitive than IgM and IgG testing for detection of SARS-CoV-2 infection.

Thus, antibody testing might play vital roles in the following settings: (1) for suspected patients at the initial visit or clinically diagnosed patients who have not been confirmed by RNA testing, the positive result of antibody increases the confidence to make a COVID-19 diagnosis; (2) for healthy close contacts who are in the quarantine period, he/she should be deemed as a probable carriers if antibody positive, then the RNA should

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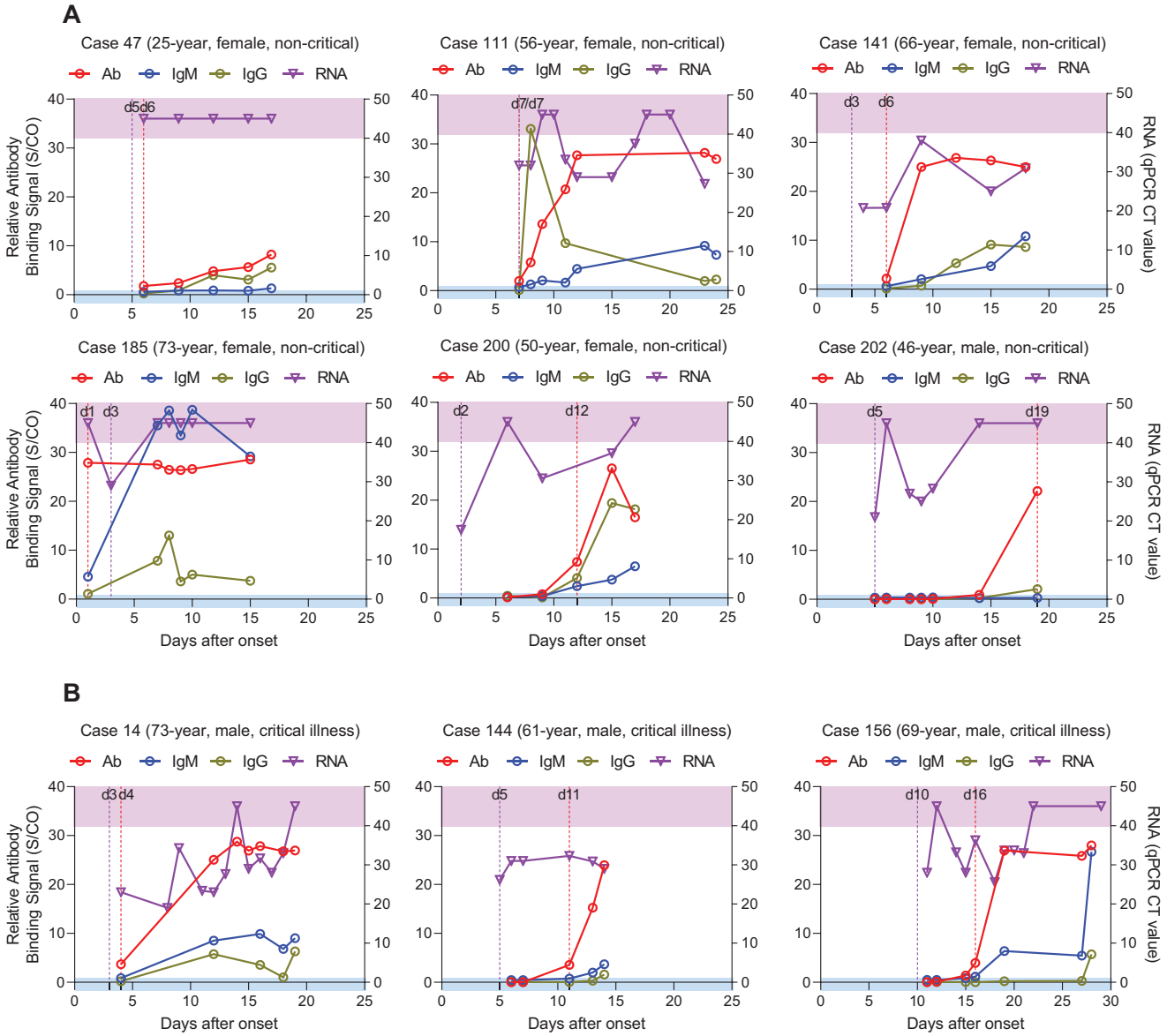


Figure 2. Dynamic profiling of viral RNA and antibodies in representative patients with COVID-19 since onset of disease. The changes in the levels of RNA in upper respiratory specimens (nasal and/or throat swabs) and antibodies (Ab, IgM, and IgG) in plasma of 9 patients are presented. Among these cases, 6 were in normal to moderate illness condition (*A*) and 3 were in critical condition (*B*). The cutoff values for antibody tests were S/CO = 1 (plotted at left *y* axis) and CT = 40 for RNA detection (plotted at right *y* axis). RNA negative samples are denoted with a CT of 45. The blue shaded area indicates the antibody seronegative zone, whereas the purple shaded area indicates undetectable RNA zone. The purple broken line indicates the first time point with detectable RNA and a red broken line indicates the first antibody seroconversion (Ab) time point. Abbreviations: Ab, total antibodies; COVID-19, coronavirus disease 2019; CT, computed tomography; Ig, immunoglobulin; qPCR, quantitative polymerase chain reaction.

be tested more frequently and the close contacts of him/her should be observed; (3) for the RNA-confirmed patient, being seropositive indicates that the specific antibodies have been induced. In addition, epidemiological studies could be conducted using immunoassays and they can play an important role in searching for potential animal hosts for SARS-CoV-2 using Ab-ELISA because the double-sandwich method is not species restricted. It has been less than 3 months since SARS-CoV-2 was first transmitted to humans, and the prevalence of antibody against SARS-CoV-2 is nearly zero. Therefore, at least during the current outbreak which is likely to continue to May or June 2020, seropositive individuals could be a probably previous infector. During this short period, the total antibody level could be considered a recent infection marker similar to IgM. Because of its higher sensitivity compared with IgM and IgG, Ab detection should be given high priority to be implemented in current clinical and public health practice. If SARS-CoV-2 becomes a common respiratory transmission pathogen in humans, such as influenzas or other less-pathogenic coronaviruses, rather than being completely eradicated such as SARS-CoV-1, the

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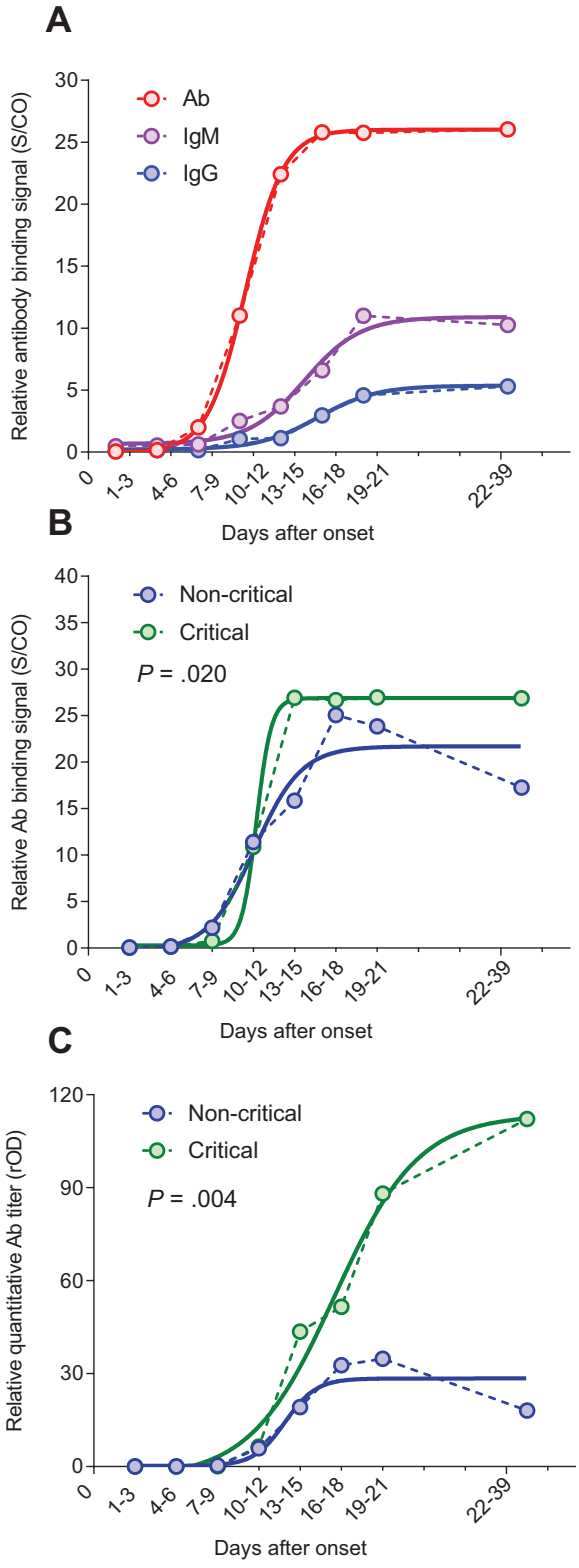


Figure 3. Average levels of antibodies against SARS-CoV-2 among patients with COVID-19 since illness onset. *A*, Comparison of the average S/CO value between Ab, IgG, and IgM. Comparison of the average S/CO value (*B*) and relative quantitative titer (*C*) of the Ab test between critical and noncritical patients. The medians of antibody detection value (S/CO for tests of Ab, IgM, and IgG, for panels *A* and *B*) and of total Ab titer (rOD, for panel *C*) of samples at the same time point since onset were used to plot the graph. Patients' samples collected from days 1–3, 4–6, 7–9, 10–12, 13–15, 16–18, 19–21, and 22–39 since illness onset were pooled for analysis. Four parameter logistic (4PL) fitting curves were used to show the rising trend of antibodies (Ab, IgG, IgM). Abbreviations: Ab, total antibodies; COVID-19, coronavirus disease 2019; Ig, immunoglobulin; rOD, relative optical density; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

serological diagnosis of acute SARS-CoV-2 infection will be more dependent on the detection of IgM in postepidemic areas in the subsequent epidemic seasons. The Ab and IgG tests could be used to understand the epidemiology of SARS-CoV-2 infection and to assist in determining the level of humoral immunity in patients. Even then, the Ab will be a more sensitive marker for sentinel monitoring of imported cases in a naive community.

In addition to the diagnosis value, our study revealed a strong positive correlation between clinical severity and Ab titer at 2 week after illness onset, for the first time in patients with COVID-19. The results suggest that a high Ab titer may be considered as a risk factor for critical illness, independently from older age, male gender, and comorbidities (Supplementary Table 2). Although the causal relation between hormonal response and illness severity is still unclear, the results indicate a possible use of a high Ab titer as a surrogate marker for worse clinical prognosis. Furthermore, it might be evidence for the possibility of antibody-dependent enhancement effects, which were commonly found in patients with SARS-CoV-1 [11, 12]. In any case, our findings suggest that the clinical meaning of the level of antibody against SARS-CoV-2 during the acute phase of infection warrants further study.

It should be noted that there are some limitations of this study. First, most of the RNA tests of the patients were based on upper respiratory tract specimens, and the positive rate may be higher in detection using lower respiratory tract specimens, such as bronchoalveolar lavage fluid and deep tracheal aspirates, which may yield higher sensitivity for RNA tests. Second, all of the patients enrolled in this study were based on the positive findings of RT-PCR using respiratory samples. Nontypical patients with lower respiratory viral load might be missed. In addition, the performance of RT-PCR depends on many factors, such as the quality and consistency of the PCR assays being used, the skill of sample collection, and the sample types. Also, RT-PCR cannot distinguish the viable replicating virus and simply residual RNA, which may affect the results. Third, we cannot evaluate the persistence of antibodies because samples were collected during the acute phase of disease. Fourth, although it has shown good specificity in healthy people, the cross-reactivity of the assay to other coronaviruses should be further assessed.

In conclusion, the findings demonstrate that antibody tests have important diagnosis value in addition to RNA tests. These findings provide strong evidence for the routine application of serological antibody assays in the diagnosis and clinical management of patients with COVID-19.

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Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

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